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1. Use dictionary to store
key -> slot no.(s1-s5) value -> status("empty" or "occupied")
2. use tuple to store parking lot information parking-name, no.of slots
3. use loop to display a menu
4. park vehicle functions all
5. remove vehicle
3.show parking status
6. // summary -----> parking name,total slots, vehicles parked
7. exit
8. Maintain a list to store vehicle no.s that entered

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In [4]: # 2. Tuple (Parking Info)
parking_info = ("Fast Parking", 5)

# 1. Dictionary (Slots)
parking_slots = {'s1': "empty", 's2': "empty", 's3': "empty", 's4': "empty", 's5': "empty"}

# 4. List (Vehicle Numbers)
vehicleslist = []

# 1. Park Vehicle
def park_vehicle():
    slot = input("Enter slot number (s1 to s5): ")

    if slot in parking_slots:
        if parking_slots[slot] == "empty":
            vehicle = input("Enter vehicle number: ")
            parking_slots[slot] = "occupied"
            vehicleslist.append(vehicle)
            print("Vehicle parked successfully!")
            summary() # Auto summary update
        else:
            print("Slot already occupied!")
    else:
        print("Invalid slot!")
```

chk slot number in dict.

<- if slot empty

<- enter VN

<- update it

<- put in list

```

# 2. Remove Vehicle
def remove_vehicle():
    slot = input("Enter slot number to remove vehicle: ")

    if slot in parking_slots:
        if parking_slots[slot] == "occupied": ← - if slot occupied(full) then we remove it
            vehicle = input("Enter vehicle number: ") ← - user enter VN

            if vehicle in vehicleslist: ← - chk input (user) vehicle in list, if it
                vehicleslist.remove(vehicle) is exist it before then we remove

                parking_slots[slot] = "empty" ← - update slot status

                print("Vehicle removed successfully!")
                summary() # Auto summary update
            else:
                print("Vehicle not found!")
        else:
            print("Slot already empty!")
    else:
        print("Invalid slot!")

# 3. Show Parking Status
def show_status():
    print("\nParking Status:") ← - use loop
    for slot in parking_slots:
        print(slot, ": ", parking_slots[slot])

# 4. Summary
def summary():
    print("\n===== Parking Summary =====")
    print("Parking Name:", parking_info[0]) ← - see tuple
    print("Total Slots:", parking_info[1])
    print("Vehicles Parked:", len(vehicleslist))

    print("\nSlot Details:") ← - use loop
    for slot in parking_slots:
        print(slot, ":", parking_slots[slot])

    print("Vehicle Numbers:", vehicleslist)

# 3. Menu Loop
while True:
    print("\n---- Parking Menu ----")
    print("1. Park Vehicle")
    print("2. Remove Vehicle")
    print("3. Show Parking Status")
    print("4. Summary")
    print("5. Exit")

    choice = input("Enter choice: ")

    if choice == "1":

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    park_vehicle()
elif choice == "2":
    remove_vehicle()
elif choice == "3":
    show_status()
elif choice == "4":
    summary()
elif choice == "5":
    print("Exiting program...")
    break
else:
    print("Invalid choice!")
```

```
---- Parking Menu ----
1. Park Vehicle
2. Remove Vehicle
3. Show Parking Status
4. Summary
5. Exit
Vehicle parked successfully!

===== Parking Summary =====
Parking Name: Fast Parking
Total Slots: 5
Vehicles Parked: 1

Slot Details:
s1 : occupied
s2 : empty
s3 : empty
s4 : empty
s5 : empty
Vehicle Numbers: ['ABC-678']

---- Parking Menu ----
1. Park Vehicle
2. Remove Vehicle
3. Show Parking Status
4. Summary
5. Exit
Vehicle parked successfully!

===== Parking Summary =====
Parking Name: Fast Parking
Total Slots: 5
Vehicles Parked: 2

Slot Details:
s1 : occupied
s2 : empty
s3 : empty
s4 : occupied
s5 : empty
Vehicle Numbers: ['ABC-678', 'AQM-766']

---- Parking Menu ----
1. Park Vehicle
2. Remove Vehicle
3. Show Parking Status
4. Summary
5. Exit

Parking Status:
s1 : occupied
s2 : empty
s3 : empty
s4 : occupied
s5 : empty
```

```
---- Parking Menu ----
1. Park Vehicle
2. Remove Vehicle
3. Show Parking Status
4. Summary
5. Exit
Vehicle removed successfully!
```

```
===== Parking Summary =====
Parking Name: Fast Parking
Total Slots: 5
Vehicles Parked: 1
```

```
Slot Details:
s1 : empty
s2 : empty
s3 : empty
s4 : occupied
s5 : empty
Vehicle Numbers: ['AQM-766']
```

```
---- Parking Menu ----
1. Park Vehicle
2. Remove Vehicle
3. Show Parking Status
4. Summary
5. Exit
Vehicle not found!
```

```
---- Parking Menu ----
1. Park Vehicle
2. Remove Vehicle
3. Show Parking Status
4. Summary
5. Exit
```

```
===== Parking Summary =====
Parking Name: Fast Parking
Total Slots: 5
Vehicles Parked: 1
```

```
Slot Details:
s1 : empty
s2 : empty
s3 : empty
s4 : occupied
s5 : empty
Vehicle Numbers: ['AQM-766']
```

```
---- Parking Menu ----
1. Park Vehicle
2. Remove Vehicle
3. Show Parking Status
4. Summary
5. Exit
Exiting program...
```

In []: